

SOME LARGE SAMPLE PROPERTIES OF HOGG'S ADAPTIVE TRIMMED MEANS

T. de Wet

Potchefstroom University

J.W.J. van Wyk

Institute for Biostatistics, M.R.C.

Key Words: Adaptive, asymptotic, robust estimation, trimmed means.

SUMMARY

In Hogg (1974) an adaptive trimmed mean was proposed to estimate the centre of a symmetric distribution; in this estimator the trimming proportion was allowed to depend on the sample through a measure of tail length. We consider a slight modification of Hogg's estimator and show that for large samples it is a.s. very close to a non-adaptive trimmed mean. The large sample distribution of the random trimming proportion is obtained and some properties of a Winsorized variance type estimator of the asymptotic variance of the adaptive trimmed means are given.

1. Introduction

Let $X_1, X_2, \dots, X_n$ be a sample of i.i.d. random variables from a continuous distribution function (df) $F(x-\theta)$, symmetric about θ , and let $X_{(1)} < X_{(2)} < \dots < X_{(n)}$ be the corresponding order statistics.

A well-known and simple robust estimator of θ is the trimmed mean, defined for $0 < \alpha < \frac{1}{2}$ as

$$m_n(\alpha) = (n - 2[n\alpha])^{-1} \sum_{i=[n\alpha]+1}^{n-[n\alpha]} X_{(i)}$$

with $[z]$ the greatest integer in z . A drawback of the non-adaptive trimmed means is that the trimming proportion α has to be fixed in advance. In the adaptive trimmed means, α is allowed to depend on the sample; e.g. Jaeckel (1971) chooses that α which minimizes an estimator of the asymptotic variance of $m_n(\alpha)$. We will follow Hogg (1974) in choosing α a function of a measure of tail length. This measure is defined for $0 < \beta < \gamma \leq \frac{1}{2}$, as

$$Q_n(\beta, \gamma) \equiv Q_n = (\bar{U}_n(\beta) - \bar{L}_n(\beta)) / (\bar{U}_n(\gamma) - \bar{L}_n(\gamma))$$

with $\bar{U}_n(\beta)$ the average of the $[n\beta]$ largest order statistics and $\bar{L}_n(\beta)$ the average of the

$[n\beta]$ smallest order statistics; likewise for $\bar{U}_n(\gamma)$ and $\bar{L}_n(\gamma)$. Hogg found that the choice $\beta = 0,2$ and $\gamma = 0,5$ makes a good test statistic for detecting tail length for distributions ranging from the normal to the double exponential.

Since α depends on the sample we denote it by α_n . We will choose $\alpha_n = K(Q_n)$ with

$$K(x) = \begin{cases} A_2 & \text{for } x > c_2 \\ \frac{A_2 - A_1}{c_2 - c_1} (x - c_2) + A_2 & \text{for } c_1 < x \leq c_2 \\ A_1 & \text{for } x \leq c_1 \end{cases}$$

with $0 < A_1 < A_2 < \frac{1}{2}$ and c_1, c_2 fixed numbers.

We then take $m_n(\alpha_n)$ as estimator of θ i.e. a trimmed mean with random trimming proportion α_n . We note that the results of the next sections continue to hold for other choices of the function K . However, the above choice is a natural and simple one to work with.

It is known that the efficiency of $m_n(\alpha)$ varies substantially as α varies through its range of possible values. In de Wet and van Wyk (1979) it was found that the efficiency of $m_n(\alpha_n)$ is rather insensitive to the choice of A_1, A_2, c_1, c_2 . In particular it was found that any choice of A_1, A_2, c_1, c_2 in the range $0,05 \leq A_1 \leq 0,15; 0,35 \leq A_2 \leq 0,45; 1,75 \leq c_1 \leq 1,80$ and $1,90 \leq c_2 \leq 1,95$ gave rather efficient estimators for distributions ranging from the Normal to the Cauchy.

In this paper some large sample properties of the adaptive trimmed means $m_n(\alpha_n)$ are investigated. In the next section we give some preliminary results needed for the main results in section 3.

2. Preliminary Results

Throughout this paper we assume $F(x - \theta)$ to be continuous and symmetric about θ ; since $m_n(\alpha_n)$ is location invariant, we will take $\theta = 0$ without loss of generality. Denote g_1, g_2, g_3 and g_4 by $F(g_1) = A_1, F(g_2) = 1 - A_1, F(g_3) = A$ and $F(g_4) = 1 - A$, with A given by (2) below. We also assume throughout that F has a positive derivative F' on $(g_1 - \epsilon_0, g_2 + \epsilon_0)$ for some $\epsilon_0 > 0$, and a bounded second derivative in some neighbourhoods of g_3 and g_4 .

For r real, let $\nu_r = \int_{-\infty}^{\infty} |x|^r dF(x)$ and denote F^{-1} by H . If they exist, let

$$(1) \quad Q(\beta, \gamma) \equiv Q = \beta^{-1} \gamma \left(\int_{1-\beta}^1 H(u) du - \int_0^\beta H(u) du \right) \left(\int_{1-\gamma}^1 H(u) du - \int_0^\gamma H(u) du \right)^{-1}$$

$$(2) \quad A = K(Q)$$

denote the "limiting" values of Q_n and α_n ; note that Q and A depend on F .

Put $U_i = F(X_i)$, $U_{(i)} = F(X_{(i)})$ and let G_n denote the empirical df of $\{U_i\}$. Let $d_n = (\log n)^{\frac{1}{2}} \delta_n$, for any sequence $\delta_n \rightarrow \infty$.

Lemma 1:

For all b , $A_1 \leq a_1 < a_2 \leq 1 - A_1$,

$$(3) \quad P(n^{\frac{1}{2}} \max_{[a_1 n] + 1 \leq i \leq [a_2 n]} |X_{(i)} - H(i/(n+1))| > \varepsilon d_n) \leq n^{-b}$$

for all $\varepsilon > 0$

Proof:

The lefthand side of (3) is bounded by

$$\sum_{[a_1 n] + 1}^{[a_2 n]} P(n^{\frac{1}{2}} |X_{(i)} - H_{in}| > \varepsilon d_n)$$

with $H_{in} = H(i/(n+1))$. It is well-known that $U_{(i)} = F(X_{(i)})$ has the same distribution as $(Z_1 + \dots + Z_i)/(Z_1 + \dots + Z_{n+1})$, with $\{Z_i\}$ i.i.d. exponential random variables with parameter 1. (See e.g. Breiman (1968), p. 285). Using this fact it follows by Bernstein's inequality that

$$P(n^{\frac{1}{2}} |X_{(i)} - H_{in}| > \varepsilon d_n) \leq n^{-b}$$

for all b , independent of $[a_1 n] + 1 \leq i \leq [a_2 n]$. The result follows from this.

Remark:

Note that lemma 1 implies

$$P(n^{-\frac{1}{2}} \sum_{i=[a_1 n] + 1}^{[a_2 n]} (X_{(i)} - H_{in}) > \varepsilon d_n) \leq n^{-b}$$

for all b .

Lemma 2:

For fixed $0 < \lambda < 1$ we have

$$P(n^{\frac{1}{2}} |G_n(\lambda) - \lambda| > \varepsilon d_n) \leq n^{-b}$$

for all b and all $\varepsilon > 0$

Proof:

This follows easily by applying Bernstein's inequality. In the next lemma we give a result on large deviations for sums of order statistics; although an iterated logarithm law for sums of order statistics follows from recent results of Wellner (1977) we also need the result of lemma 3.

Lemma 3:

Suppose $v_3 < \infty$ (in addition to the previous stated conditions on F). For $0 \leq \delta < \lambda \leq 1$ we have for all $\varepsilon > 0$

$$(4) \quad P(n^{\frac{1}{2}} |n^{-1} \sum_{i=[n\delta]+1}^{[n\lambda]} X_{(i)} - \int_{\delta}^{\lambda} H(u) du| > \varepsilon d_n) = o(n^{-\frac{1}{2}} (\log n)^{-2})$$

Proof:

We will only show that

$$(5) \quad P(n^{\frac{1}{2}} |n^{-1} \sum_{i=1}^{[n\delta]} X_{(i)} - \int_0^{\delta} H(u) du| > \varepsilon d_n) = o(n^{-\frac{1}{2}} (\log n)^{-2})$$

The remainder of the proof of the lemma follows in an analogous way.

Note that without loss of generality we may take δ in (5) as small as we wish. Now, using the probability integral transformation it follows that

$$\begin{aligned} R_n &\equiv n^{-1} \sum_{i=1}^{[n\delta]} X_{(i)} - \int_0^{\delta} H(u) du \\ &= \int_0^1 H(u) I(G_n(u) \leq \delta) dG_n(u) - \int_0^1 H(u) I(u \leq \delta) du \end{aligned}$$

with $I(\cdot)$ the indicator function. Let $a_n = (\delta_n \log n)^{\frac{1}{2}}$ so that $a_n = o(d_n)$. Then

$$\begin{aligned} R_n &= \int_0^1 H(u) I(u \leq \delta) d(G_n(u) - u) - \int_0^{\delta} H(u) I(G_n(u) > \delta + n^{-\frac{1}{2}} a_n) dG_n(u) \\ &\quad - \int_0^{\delta} H(u) I(\delta < G_n(u) \leq \delta + n^{-\frac{1}{2}} a_n) dG_n(u) \\ &\quad + \int_{\delta}^1 H(u) I(G_n(u) \leq \delta - n^{-\frac{1}{2}} a_n) dG_n(u) \\ &\quad + \int_{\delta}^1 H(u) I(\delta - n^{-\frac{1}{2}} a_n < G_n(u) \leq \delta) dG_n(u) \end{aligned}$$

$$= R_{1n} - R_{2n} - R_{3n} + R_{4n} + R_{5n} \quad (\text{say})$$

Since

$$R_{1n} = n^{-1} \sum_{i=1}^n (H(U_i)I(U_i \leq \delta) - EH(U_i)I(U_i \leq \delta)).$$

is an average of i.i.d. random variables, it follows from Theorem 5 of Michel (1976) that

$$P(n^{\frac{1}{2}} |R_{1n}| > \varepsilon d_n) = o(n^{-\frac{1}{2}} (\log n)^{-2})$$

Since G_n is non-decreasing, it follows that

$$|R_{2n}| \leq I(G_n(\delta) > \delta + n^{-\frac{1}{2}} a_n) \int_0^1 |H(u)| dG_n(u).$$

From lemma 2 it follows that for n large enough

$$P(n^{\frac{1}{2}} I(G_n(\delta) > \delta + n^{-\frac{1}{2}} a_n) > \varepsilon d_n) \leq P(G_n(\delta) > \delta + n^{-\frac{1}{2}} a_n) \leq n^{-b}, \text{ for all } b.$$

Since

$$\int_0^1 |H(u)| dG_n(u) = n^{-1} \sum_{i=1}^n |H(U_i)|$$

is an average of i.i.d. random variables, Theorem 5 of Michel (1976) can be applied again, and we conclude

$$P(n^{\frac{1}{2}} |R_{2n}| > \varepsilon d_n) = o(n^{-\frac{1}{2}} (\log n)^{-2}).$$

Let G_n^{-1} be the usual inverse of G_n ; noting that

$$G_n^{-1}(v) = U_{(j)} \text{ for } (j-1)/n < v \leq j/n,$$

$|H(u)|$ is decreasing on $(0, \frac{1}{2})$, and taking $\delta < \frac{1}{2}$ we have

$$\begin{aligned} |R_{3n}| &\leq \int_0^\delta |H(u)| I(\delta < G_n(u) \leq \delta + n^{-\frac{1}{2}} a_n) dG_n(u) \\ &\leq \int_0^\delta |H(u)| I(G_n^{-1}(\delta) \leq u \leq G_n^{-1}(\delta + n^{-\frac{1}{2}} a_n)) dG_n(u) \\ &= I(G_n^{-1}(\delta) \leq \delta) \int_0^\delta |H(u)| I(G_n^{-1}(\delta) \leq u \leq G_n^{-1}(\delta + n^{-\frac{1}{2}} a_n)) dG_n(u) \end{aligned}$$

$$\begin{aligned}
&\leq |H(G_n^{-1}(\delta))| |I(G_n^{-1}(\delta))| \leq \delta n^{-1} \sum_{j=1}^n |I(G_n^{-1}(\delta))| \leq U_{(j)} \leq G_n^{-1}(\delta + n^{-\frac{1}{2}} a_n) \\
&\leq |X_{([n\delta])}| n^{-1} \sum_j I(U_{([n\delta])} \leq U_{(j)} \leq U_{([n(\delta + n^{-\frac{1}{2}} a_n)] + 1)}) \\
&\leq (|X_{([n\delta])}| - H(\delta)) + |H(\delta)| (n^{-\frac{1}{2}} a_n + 3n^{-1}).
\end{aligned}$$

Using a special case of Lemma 1 and the fact that $a_n = o(d_n)$ it follows that

$$P(n^{\frac{1}{2}} |R_{3n}| > \varepsilon d_n) \leq n^{-b}$$

for all b .

R_{4n} and R_{5n} can be handled in a similar way as R_{2n} and R_{3n} respectively. We conclude that

$$P(n^{\frac{1}{2}} |R_n| > \varepsilon d_n) = o(n^{-\frac{1}{2}} (\log n)^{-2})$$

and the lemma follows.

In addition to the previous conditions, we need the following condition on F for our next result.

(F1) $H = F^{-1}$ is continuous in the points $\beta, 1-\beta, \gamma, 1-\gamma$.

Remark:

Note that if $A_1 \leq \beta$ then $1-\beta \leq 1-A_1$ so that (F1) is covered by our previous conditions on F .

Let

$$S_1(v) = v^{-1} \int_0^v H(u) du; S_2(v) = v^{-1} \int_v^1 H(u) du$$

so that

$$Q = (S_2(\beta) - S_1(\beta)) / (S_2(\gamma) - S_1(\gamma))$$

Also let

$$d_1 = \beta^{-1} (S_2(\gamma) - S_1(\gamma))^{-1} - \gamma^{-1} (S_2(\beta) - S_1(\beta)) (S_2(\gamma) - S_1(\gamma))^{-2}$$

$$d_2 = -\gamma^{-1} (S_2(\beta) - S_1(\beta)) (S_2(\gamma) - S_1(\gamma))^{-2}$$

$$J(u) = \begin{cases} -d_1 & \text{for } 0 \leq u < \beta \\ -d_2 & \text{for } \beta \leq u < \gamma \\ 0 & \text{for } \gamma \leq u < 1 - \gamma \\ d_2 & \text{for } 1 - \gamma \leq u < 1 - \beta \\ d_1 & \text{for } 1 - \beta \leq u < 1 \end{cases}$$

We have the following:

Lemma 4:

(i) If $\nu_r < \infty$ for some $r > 0$ then

$$Q_n = Q + o(n^{-\frac{1}{2}} \log_2 n) \text{ a.s.}$$

(ii) If $\nu_1 < \infty$ and (F1) holds, then $\sigma^2(J, F) < \infty$

implies

$$n^{\frac{1}{2}}(Q_n - Q) \xrightarrow{D} N(0, \sigma^2(J, F)) \text{ with}$$

$$(6) \quad \sigma^2(J, F) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} J(F(x))J(F(y))(F(\min(x, y)) - F(x)F(y))dx dy$$

(iii) If $\nu_3 < \infty$ then

$$P(n^{\frac{1}{2}}|Q_n - Q| > \varepsilon d_n) = o(n^{-\frac{1}{2}} (\log n)^{-2}).$$

Proof:

It follows directly that

$$(7) \quad Q_n - Q = [(t_{2n}(\beta) - t_{1n}(\beta))t(\gamma) - (t_{2n}(\gamma) - t_{1n}(\gamma))t(\beta)]t_n(\gamma)^{-1}t(\gamma)^{-1}$$

$$= q_n(\beta, \gamma) t_n(\gamma)^{-1}t(\gamma)^{-1} \quad (\text{say})$$

with

$$t_{1n}(v) = [nv]^{-1} \sum_{i=1}^{[nv]} X_{(i)} - S_1(v) = S_{1n}(v) - S_1(v) \quad (\text{say})$$

$$t_{2n}(v) = [nv]^{-1} \sum_{n-[nv]+1}^n X_{(i)} - S_2(v) = S_{2n}(v) - S_2(v) \quad (\text{say})$$

$$t(v) = S_2(v) - S_1(v) \text{ and } t_n(v) = S_{2n}(v) - S_{1n}(v)$$

(i) Using Theorem 4 of Wellner (1977) it follows that

$$\begin{aligned} Q_n - Q &= q_n(\beta, \gamma) t(\gamma)^{-2} + O(n^{-1} \log_2 n) \text{ a.s.} \\ &= O(n^{-\frac{1}{2}} \log_2 n) \text{ a.s.} \end{aligned}$$

(ii) This follows directly from David (1970), Theorem 9.6, p. 217.

(iii) To prove (iii) we need the following, which follows quite easily. Let Y_1 and Y_2 be random variables and a, b, c real numbers with $a, b > 0$, then

$$(8) \quad P(Y_1 > aY_2/b) \leq P(Y_1 > a(1-c)) + P(|Y_2 - b| > bc)$$

Choosing $a = \varepsilon d_n t(\gamma)^2$, $b = t(\gamma)$ and $c = n^{-\frac{1}{2}} d_n$ in (8) it follows from (7) on noting that $t_n(\gamma), t(\gamma) > 0$ a.s., that

$$\begin{aligned} P(n^{\frac{1}{2}} |Q_n - Q| > \varepsilon d_n) &= P(n^{\frac{1}{2}} |q_n(\beta, \gamma)| > \varepsilon d_n t_n(\gamma) t(\gamma)) \\ &\leq P(n^{\frac{1}{2}} |q_n(\beta, \gamma)| > \varepsilon t(\gamma)^2 d_n (1 - n^{-\frac{1}{2}} d_n)) + P(n^{\frac{1}{2}} |t_n(\gamma) - t(\gamma)| > t(\gamma) d_n) \\ &= o(n^{-\frac{1}{2}} (\log n)^{-2}) \end{aligned}$$

by repeated application of lemma 3. The desired result follows.

3. Main Results

In this section the main results are given. We first show that our random trimming proportion is close to a nonrandom number. Let

$$(9) \quad h_Q(x) = x[D_-(K(Q))I(x < 0) + D_+(K(Q))I(x \geq 0)]$$

with D_- and D_+ denoting left and right derivatives. Note that $h_Q(cx) = ch_Q(x)$ for $c > 0$. We still assume the stated conditions on F to hold.

Theorem 1:

(i) If $\nu_r < \infty$ for some $r > 0$, then $\alpha_n = A + O(n^{-\frac{1}{2}} \log_2 n)$ a.s.

(ii) If $\nu_1 < \infty$ and (F1) holds, then $\sigma^2(J, F) < \infty$ implies

$$(10) \quad n^{\frac{1}{2}} (\alpha_n - A) \xrightarrow{D} h_Q(Z)$$

with Z a $N(0, \sigma^2(J, F))$ random variable, with $\sigma^2(J, F)$ given by (6).

(iii) If $v_3 < \infty$ then

$$P(n^{\frac{1}{2}} |\alpha_n - A| > \epsilon d_n) = o(n^{-\frac{1}{2}} (\log n)^{-2}), \text{ with } d_n \text{ as before.}$$

Proof:

(1) and (iii) follow immediately from lemma 4 on noting that

$$|\alpha_n - A| \leq |Q_n - Q| (A_2 - A_1) (c_2 - c_1)^{-1}$$

We now prove (ii). We can write

$$\begin{aligned} n^{\frac{1}{2}} (\alpha_n - A) &= n^{\frac{1}{2}} h_Q(Q_n - Q) + n^{\frac{1}{2}} (K(Q_n) - K(Q) - h_Q(Q_n - Q)) \\ &= h_Q(n^{\frac{1}{2}} (Q_n - Q)) + n^{\frac{1}{2}} \Delta_n g(\Delta_n) \end{aligned}$$

$$\text{with } \Delta_n = Q_n - Q, \text{ i.e. } Q_n = \Delta_n + Q$$

$$g(x) = (K(x + Q) - K(Q) - h_Q(x))x^{-1}$$

By lemma 4 (ii) and continuity of h_Q we have

$$h_Q(n^{\frac{1}{2}} (Q_n - Q)) \xrightarrow{D} h_Q(Z)$$

with Z a $N(0, \sigma^2(J, F))$ random variable.

It follows easily that

$$\lim_{x \rightarrow 0} g(x) = 0$$

and since $\Delta_n = o_p(1)$, $n^{\frac{1}{2}} \Delta_n = O_p(1)$, we have $n^{\frac{1}{2}} \Delta_n g(\Delta_n) = o_p(1)$ as $n \rightarrow \infty$. The desired result follows.

Remark:

Note that $h_Q(Z)$ is a normal random variable for all Q except $Q = c_1, c_2$; for $Q < c_1$ or $Q > c_2$, $h_Q(Z)$ is degenerate at 0. The nonnormality of $h_Q(Z)$ in c_1 and c_2 is due to the fact that the left and right derivatives of K differ in these points.

Theorem 2:

If $v_r < \infty$ for some $r > 0$, then

$$m_n(\alpha_n) = m_n(A) + O(n^{-1}(\log n)^{\frac{1}{2}} (\log_2 n)^2) \text{ a.s.}$$

Proof:

Let

$$\begin{aligned} V_n &= \left(\sum_{i=[n\alpha_n]+1}^{[nA]} X_{(i)} + \sum_{i=n-[nA]+1}^{n-[n\alpha_n]} X_{(i)} \right) I([n\alpha_n] < [nA]) \\ &- \left(\sum_{i=[nA]+1}^{[n\alpha_n]} X_{(i)} + \sum_{i=n-[n\alpha_n]+1}^{n-[nA]} X_{(i)} \right) I([n\alpha_n] > [nA]) \\ &= \left(\sum_{i=[n\alpha_n]+1}^{[nA]} (X_{(i)} - H_{in}) + \sum_{i=n-[nA]+1}^{n-[n\alpha_n]} (X_{(i)} - H_{in}) \right) I([n\alpha_n] < [nA]) \\ &- \left(\sum_{i=[nA]+1}^{[n\alpha_n]} (X_{(i)} - H_{in}) + \sum_{i=n-[n\alpha_n]+1}^{n-[nA]} (X_{(i)} - H_{in}) \right) I([n\alpha_n] > [nA]) \end{aligned}$$

by the symmetry of F , with $H_{in} = H(i/(n+1))$

Then

$$m_n(\alpha_n) - m_n(A) = R_{1n} + R_{2n} \quad (\text{say})$$

with

$$R_{1n} = 2([n\alpha_n] - [nA])(n - 2[n\alpha_n])^{-1} m_n(A),$$

$$R_{2n} = (n - 2[n\alpha_n])^{-1} V_n$$

We have $|[n\alpha_n] - [nA]| \leq n|\alpha_n - A| + 1$, $(n - 2[n\alpha_n])^{-1} \leq n^{-1}(1 - 2A_2)^{-1}$ a.s., since

$\alpha_n \leq A_2$ a.s. From Theorem 1 of de Wet and Venter (1974) we have

$$m_n(A) = O(n^{-\frac{1}{2}} (\log_2 n)^{\frac{1}{2}}) \text{ a.s.}$$

Using this and Theorem 1 (i) we conclude that $R_{1n} = O(n^{-1} (\log_2 n)^{3/2})$ a.s. Note that

$|R_{2n}| \leq n^{-1} (1 - 2A_2)^{-1} |V_n|$ a.s.; we will prove that $V_n = O(d_n \log_2 n)$ a.s.

$$\text{Let } V_{1n} \equiv \sum_{i=[n\alpha_n]+1}^{[nA]} (X_{(i)} - H_{in}) I([n\alpha_n] < [nA])$$

Using lemma 1, the Borel-Cantelli lemma and Theorem 1 (i) it follows that

$$\begin{aligned}
|V_{1n}| &\leq |[nA] - [n\alpha_n]| \max_{[nA_1] \leq i \leq [nA]} |X_{(i)} - H_{in}| \\
&\leq (n|\alpha_n - A| + 1) \max_{[nA_1] \leq i \leq [nA]} |X_{(i)} - H_{in}| \\
&= O(d_n \log_2 n) \text{ a.s.}
\end{aligned}$$

The other terms in V_n can be handled in a similar fashion and we conclude that $R_{2n} = O(n^{-1} d_n \log_2 n)$ a.s. The theorem follows by choosing $\delta_n = \log_2 n$.

Remarks:

1. This result says that the adaptive trimmed mean is a.s. very close to a non-adaptive trimmed mean for every F (satisfying certain conditions). Thus if A (a function of F) is a reasonable trimming proportion for F we would expect $m_n(\alpha_n)$ to be a reasonably good estimator since for every F , $m_n(A)$ is reasonably good.

2. Using the above result and Theorem 1 of de Wet and Venter (1974) it follows that

$$m_n(\alpha_n) - \theta = n^{-1} \sum_{i=1}^n Y_i + O(n^{-1} (\log n)^{\frac{1}{2}} (\log_2 n)^2) \text{ a.s.}$$

with $\{Y_i\}$ i.i.d. random variables defined there. From this it follows e.g. that an iterated logarithm law holds for $m_n(\alpha_n)$ and that $n^{\frac{1}{2}} (m_n(\alpha_n) - \theta)$ is asymptotically $N(0, \sigma^2(A))$ distributed with

$$\sigma^2(A) = (1-2A)^{-2} \left[\int_{g_3}^{g_4} (x - \theta)^2 dF(x) + A(g_3 - \theta)^2 + A(g_4 - \theta)^2 \right]$$

with $F(g_3) = A, F(g_4) = 1 - A$.

3. It follows directly that the influence curve of $m_n(\alpha_n)$ is the same as that of $m_n(A)$ and hence is bounded and continuous.

We now show that a Berry-Esseen type result holds for $m_n(\alpha_n)$.

Theorem 3:

Let

$$F_n(x) = P(n^{\frac{1}{2}} (m_n(\alpha_n) - \theta) \leq x \mid \sigma(A))$$

If $\nu_3 < \infty$, then

$$\sup_{-\infty < x < \infty} |F_n(x) - \Phi(x)| = O(n^{-\frac{1}{2}} (\log n)^2)$$

with Φ the $N(0;1)$ df.

Proof:

Without loss of generality we again take $\theta = 0$. From the proof of theorem 2 above, we have

$$m_n(\alpha_n) - m_n(A) = R_{1n} + R_{2n}$$

With $\rho_n = n^{-\frac{1}{2}} (\log n)^2$, it follows from Theorem 1 of de Wet (1976) that

$$(11) \quad P(n^{\frac{1}{2}} m_n(A) \leq x \sigma(A) + \rho_n) \leq \Phi(x) + \sigma(A)^{-1} (2\pi)^{-\frac{1}{2}} \rho_n + B\rho_n$$

with B independent of x , and

$$(12) \quad P(n^{\frac{1}{2}} m_n(A) \leq x \sigma(A) - \rho_n) \geq \Phi(x) - \sigma(A)^{-1} (2\pi)^{-\frac{1}{2}} \rho_n - B\rho_n$$

Using the inequality $1 - \Phi(x) < x^{-1} \phi(x)$, for $x > 0$, with ϕ the $N(0;1)$ density, we also have

$$(13) \quad P(n^{\frac{1}{2}} |m_n(A)| > \log n) = O(\rho_n)$$

Since $|R_{1n}| \leq 2(1-A_2)^{-1} (|\alpha_n - A| + n^{-1}) |m_n(A)|$,

it follows from Theorem 1 (iii) and (13) that

$$\begin{aligned} (14) \quad P(n |R_{1n}| > \varepsilon (\log n)^2) &\leq P(2(1-2A_2)^{-1} n^{\frac{1}{2}} (|\alpha_n - A| + n^{-1}) > \varepsilon \log n) \\ &\quad + P(n^{\frac{1}{2}} |m_n(A)| > \log n) \\ &= O(\rho_n). \end{aligned}$$

With V_n, V_{1n} as in the proof of Theorem 2, we have

$$|R_{2n}| \leq n^{-1} (1-2A_2)^{-1} |V_n|$$

and

$$|V_{1n}| \leq (n|\alpha_n - A| + 1) \max_{[nA_1] \leq i \leq [nA]} |X_{(i)} - H_{in}|$$

Using lemma 1 and Theorem 1 (iii) it follows directly that

$$P(|V_{1n}| > \varepsilon(\log n)^2) = o(n^{-\frac{1}{2}}(\log n)^{-2})$$

The other terms in V_n can be handled in a similar way and we conclude that

$$(15) \quad P(|R_{2n}| > \varepsilon(\log n)^2) = o(n^{-\frac{1}{2}}(\log n)^{-2}).$$

Using (11), (12), (14) and (15) it follows that

$$\begin{aligned} F_n(x) &= P(n^{\frac{1}{2}}m_n(A) + n^{\frac{1}{2}}R_{1n} + n^{\frac{1}{2}}R_{2n} \leq x\sigma(A) + \rho_n - \rho_n) \\ &\leq P(n^{\frac{1}{2}}m_n(A) \leq x\sigma(A) + \rho_n) + P(n^{\frac{1}{2}}R_{1n} \leq -\frac{1}{2}\rho_n) + P(n^{\frac{1}{2}}R_{2n} \leq -\frac{1}{2}\rho_n) \\ &\leq \Phi(x) + r_n \end{aligned}$$

and

$$\begin{aligned} F_n(x) &\geq P(n^{\frac{1}{2}}m_n(A) \leq x\sigma(A) - \rho_n) - P(n^{\frac{1}{2}}R_{1n} \geq \frac{1}{2}\rho_n) - P(n^{\frac{1}{2}}R_{2n} \geq \frac{1}{2}\rho_n) \\ &\geq \Phi(x) - r_n \end{aligned}$$

with

$$\begin{aligned} r_n &= \sigma(A)^{-1}(2\pi)^{-\frac{1}{2}}\rho_n + B\rho_n + P(n^{\frac{1}{2}}|R_{1n}| \geq \frac{1}{2}\rho_n) + P(n^{\frac{1}{2}}|R_{2n}| \geq \frac{1}{2}\rho_n) \\ &= O(\rho_n), \end{aligned}$$

independent of x . The theorem follows.

Remark:

The rate obtained in Theorem 3 is not best possible although very close to it; however the rate of $O(n^{-\frac{1}{2}})$, which holds for the usual trimmed mean, seems out of reach using the present method. To obtain this one would have to go back to the usual characteristic function methods which are very difficult to use in the present situation.

A well-known estimator of the asymptotic variance $\sigma^2(\alpha)$ of the trimmed mean $m_n(\alpha)$ is the Winsorized variance

$$s_n^2(\alpha) = (1-2\alpha)^{-2} [n^{-1} \sum_{i=[n\alpha]+1}^{n-[n\alpha]} (X_{(i)} - m_n(\alpha))^2 + \alpha(X_{([n\alpha]+1)} - m_n(\alpha))^2 + \alpha(X_{(n-[n\alpha])} - m_n(\alpha))^2]$$

$$+ \alpha(X_{(n-[n\alpha])} - m_n(\alpha))^2],$$

see e.g. Jaeckel (1971). Since the asymptotic variance of $m_n(\alpha_n)$ is $\sigma^2(A)$, a natural estimator of $\sigma^2(A)$ is $s_n^2(\alpha_n)$. It follows directly from lemma 2 of Jaeckel (1971) (under the conditions stated on F) that $s_n^2(\alpha_n)$ is a consistent estimator of $\sigma^2(A)$; actually $s_n^2(\alpha_n) = \sigma^2(A) + O_p(n^{-1/2})$. At first, one would expect the same type of result as in Theorem 2 to hold for $s_n^2(\alpha_n)$ and $s_n^2(A)$. This, however, is not the case as the next theorem shows.

Theorem 4:

If $v_r < \infty$ for some $r > 0$, then

$$s_n^2(\alpha_n) = s_n^2(A) + h(A)(\alpha_n - A) + O(n^{-3/4} \log n) \text{ a.s.}$$

with

$$\begin{aligned} h(A) &\equiv 4(1-2A)^{-2} ((1-2A)\sigma^2(A) + AH(A)H'(A)) \\ &= \frac{d}{dA} \sigma^2(A). \end{aligned}$$

Proof:

We will only give an outline of the proof; the details can be supplied in a straightforward way. We again take $\theta = 0$ without loss of generality. Let

$$s_{1n}^2(\alpha) = (1-2\alpha)^{-2} [n^{-1} \sum_{[n\alpha]+1}^{n-[n\alpha]} X_{(i)}^2 + \alpha X_{([n\alpha]+1)}^2 + \alpha X_{(n-[n\alpha])}^2]$$

It then follows easily using theorem 2 that

$$s_n^2(\alpha_n) - s_{1n}^2(\alpha_n) = O(n^{-3/4} \log n) \text{ a.s.}$$

and

$$s_n^2(A) - s_{1n}^2(A) = O(n^{-3/4} \log n) \text{ a.s.}$$

so we need only consider $s_{1n}^2(\alpha_n)$ and $s_{1n}^2(A)$.

From Bahadur (1966) it follows that

$$X_{([nA]+1)} = H(A) + (A - F_n^*(H(A)))H'(A) + O(n^{-\frac{3}{4}} \log n)$$

a.s., with F_n^* the empirical df of $X_1, \dots, X_n$, and a similar result for $X_{(n-[nA])}$. Lemma 3 of Bahadur (1966) can be adapted in a straightforward way to yield

$$X_{([n\alpha_n]+1)} = H(A) + (\alpha_n - F_n^*(H(A)))H'(A) + O(n^{-\frac{3}{4}} \log n)$$

a.s., since by theorem 1, $n\alpha_n = nA + O(n^{\frac{1}{2}} \log_2 n)$ a.s.; likewise for $X_{(n-[n\alpha_n])}$.

Using these representations, Theorem 1 and the idea of the proof of Theorem 2 it follows, with

$$H_{in} = H(i/(n+1)),$$

that

$$\begin{aligned} s_n^2(\alpha_n) - s_n^2(A) &= (1-2A)^{-2} \left[(n-1) \sum_{[n\alpha_n]+1}^{[nA]} H_{in}^2 + n^{-1} \sum_{n-[n\alpha_n]+1}^{n-[nA]} H_{in}^2 \right] I([n\alpha_n] < [nA]) \\ &\quad - (n-1) \sum_{[nA]+1}^{[n\alpha_n]} H_{in}^2 + n^{-1} \sum_{n-[n\alpha_n]+1}^{n-[nA]} H_{in}^2 \Big] I([n\alpha_n] > [nA]) \\ &\quad + 4(1-2A)^{-1} (\alpha_n - A) n^{-1} \sum_{[nA]+1}^{n-[nA]} H_{in}^2 + 2AH(A) (X_{([n\alpha_n]+1)} - X_{([nA]+1)}) \\ &\quad + (1-2A)^{-1} (1+2A)H(A)^2 (\alpha_n - A) + 2AH(1-A) (X_{(n-[n\alpha_n])} \\ &\quad - X_{(n-[nA])}) + (1-2A)^{-1} (1+2A)H(1-A)^2 (\alpha_n - A) \\ &\quad + O(n^{-\frac{3}{4}} \log n) \text{ a.s.} \end{aligned}$$

Since each sum in this expression approximates the corresponding integral to $O(n^{-1})$ a.s., it follows by using the above Bahadur representations that

$$\begin{aligned} s_n^2(\alpha_n) - s_n^2(A) &= (1-2A)^{-2} \left[\int_{\alpha_n}^A H(u)^2 du + \int_{1-A}^{1-\alpha_n} H(u)^2 du \right] I([n\alpha_n] < [nA]) \\ &\quad - \left(\int_A^{\alpha_n} H(u)^2 du + \int_{1-\alpha_n}^{1-A} H(u)^2 du \right) I([n\alpha_n] > [nA]) \end{aligned}$$

$$\begin{aligned}
& + 4(1-2A)^{-1} \int_A^{1-A} H(u)^2 du (\alpha_n - A) + 4AH(A)H'(A)(\alpha_n - A) \\
& + 2(1-2A)^{-1} (1+2A)H(A)^2 (\alpha_n - A) + O(n^{-\frac{3}{4}} \log n) \text{ a.s.}
\end{aligned}$$

Applying a Taylor expansion to second order to each of the integrals above, adding terms and using Theorem 1 the desired result follows.

Remarks:

1. Intuitively, $h(A)(\alpha_n - A)$ is the first order term in a Taylor development of $s_n^2(\alpha_n)$ about A . In the case of $m_n(\alpha_n)$ this first order term was absorbed into the remainder.

2. Since

$$n^{\frac{1}{2}}(s_n^2(\alpha_n) - \sigma^2(A)) = n^{\frac{1}{2}}(s_n^2(A) - \sigma^2(A)) + h(A)n^{\frac{1}{2}}(\alpha_n - A) + o_p(1),$$

from Theorem 1 and Theorem 2 of de Wet and Venter (1974), the left hand side to have a limiting distribution. Using the method of Shorack (1972) this can be shown to be the case; more precisely, its limiting distribution is the distribution of the random variable

$$\begin{aligned}
& 2(1-2A)^{-2} \left[\int_A^{1-A} H(u)H'(u)B(u)du + AH(A)H'(A)(B(A) - B(1-A)) \right] \\
& + h(A)h_Q \left(\int_0^1 J(u)H'(u)B(u)du \right),
\end{aligned}$$

with $B(u)$ the Brownian bridge process. Note that this random variable has a normal distribution for $Q \neq c_1, c_2$ and a non-normal distribution for $Q = c_1, c_2$. We omit the rather lengthy details.

3. In de Wet and van Wyk (1979) it was found that $m_n(\alpha_n)$ possesses good efficiency and robustness properties. $m_n(\alpha_n)$ and $s_n(\alpha_n)$ were also used to construct confidence intervals for θ .

REFERENCES

- BAHADUR, R.R. (1966). A Note on Quantiles in Large Samples. *Ann. Math. Statist.* 37, 577-580.
- BREIMAN, L. (1968). *Probability*. Addison-Wesley, Reading, Mass.
- DAVID, H.A. (1970). *Order Statistics*. John Wiley and Sons. New York.

- DE WET, T and VENTER, J.H. (1974). An Asymptotic Representation of Trimmed Means with Applications. *S. Afr. Statist. J.* 8, 127-134.
- DE WET, T. (1976). Berry-Esseen Results for the Trimmed Mean. *S. Afr. Statist. J.* 10, 77-96.
- DE WET, T and VAN WYK, J.W.J. (1979). Efficiency and Robustness of Hogg's Adaptive Trimmed Means. *Comm. Statist.* A8.
- HOGG, R.V. (1974). Adaptive Robust Procedures: A Partial Review and Some Suggestions for Future Applications and Theory. *J. Amer. Statist. Assoc.* 69, 909-927.
- JAECKEL, L.A. (1971). Some Flexible Estimates of Location. *Ann. Math. Statist.* 42, 1540-1552.
- MICHEL, R. (1976). Nonuniform Central Limit Bounds with Applications to Probabilities of Deviations. *Ann. Prob.* 4, 102-106.
- SHORACK, G. (1972). Functions of Order Statistics. *Ann. Math. Statist.* 43, 412-427.
- WELLNER, Jon A. (1977). A Law of the Iterated Logarithm for Functions of Order Statistics. *Ann. Statist.* 5, 481-494.

Manuscript received 1978-05-10.